STAT A411/611 Class Notes

Instructor: Rieken Venema **Office:** SSB 158E
Office hours: MWF 12:00–12:45 **Class:** 2:30–3:45

Plan and Review

Today: Section 3.3.
Wednesday: Sections 3.4–3.8.
Last Wednesday: Sections 3.1 and 3.2.

Review: Discrete Random Variables

Perform an experiment and let X be a random variable. Let $\mathcal{X}$ denote the set of possible values of X. For a **discrete** random variable, $\mathcal{X}$ is a countable set and its values can be listed. For example:

$$\{1, 2, 3\}, \qquad \{0, 1, 2, 3, \ldots\}.$$

The **probability mass function (pmf)** is

$$\boxed{p(x) = P(X = x), \qquad x \in \mathcal{X}.}$$

A valid pmf satisfies both

$$\boxed{\sum_{x \in \mathcal{X}} p(x) = 1}, \qquad \boxed{p(x) \geq 0 \quad \forall x \in \mathcal{X}.}$$

The **cumulative distribution function (cdf)** is

$$\boxed{F(x) = P(X \leq x), \qquad -\infty < x < \infty.}$$

For $0 \leq a$, the geometric sequence and series formulas are

$$\{a^0, a^1, a^2, \ldots\}, \qquad \sum_{k=0}^{\infty} a^k = \begin{cases} \dfrac{1}{1-a}, & 0 \leq a < 1, \\ \infty, & a \geq 1. \end{cases}$$

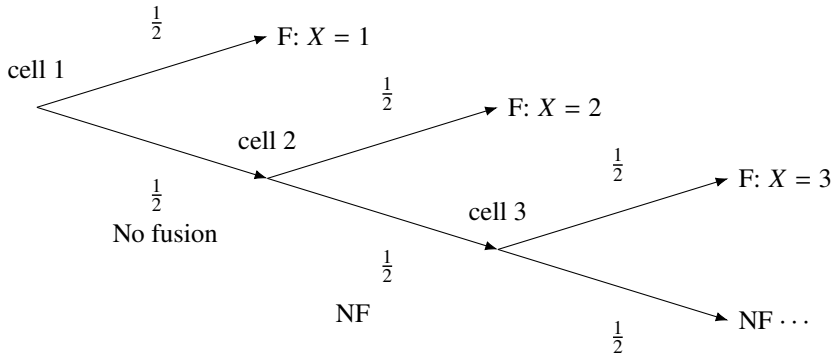
Exercises: 2, 4, 6, 8, 10, 12.

Review Example: Cells Exposed Until Fusion

Each exposed cell produces fusion with probability $1/2$. Let

$$X = \text{the number of cells exposed until a fusion takes place.}$$

The first three cells give the stopping tree below.



Find the pmf, check that it is valid, and find the cdf and related probabilities.

3 Chapter 3 - Discrete Random Variables and Their Probability Distributions

Section 3.3 - Expected Values

Let X be a discrete random variable and let $h(x)$ be a function of its possible values. The expectation formulas below are used when the relevant expectations exist; the finite-variance formulas require a finite second moment.

Definition. The **expected value**, or **mean value**, of $h(X)$ is defined by

$$\boxed{E[h(X)] = \sum_{x \in \mathcal{X}} h(x)p(x) = \sum_{x \in \mathcal{X}} h(x)P(X = x).} \qquad (*)$$

Case 1: $h(x) = x$ **- The Mean of** X

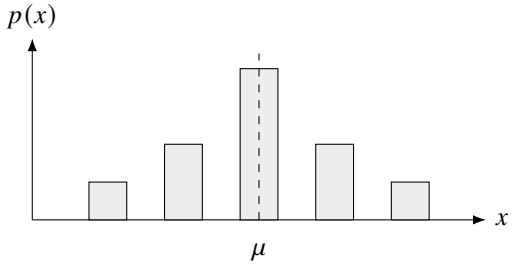
Putting $h(x) = x$ in $(*)$ gives

$$\boxed{E[X] = \sum_{x \in \mathcal{X}} x\, P(X = x).}$$

This is the **expected value** of the random variable X and is also denoted by

$$\boxed{\mu = E[X].}$$

The more likely it is that X takes a particular value x, the more weight that value receives in the sum. Thus $E[X]$ is a **weighted average** and is used as a measure of the **center** of the probability distribution of X.



The mean locates the center of the probability distribution.

Example 1: Equal Probabilities

Suppose

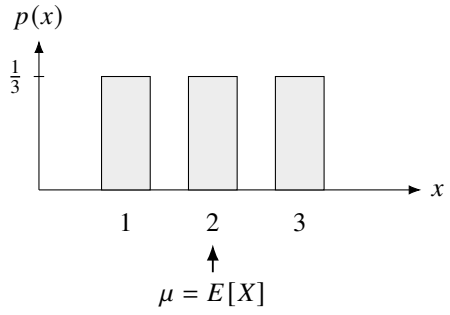
$$\mathcal{X} = \{1, 2, 3\}, \qquad p(1) = p(2) = p(3) = \frac{1}{3}.$$

The probabilities sum to one:

$$\frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1.$$

Then

$$E[X] = 1\left(\frac{1}{3}\right) + 2\left(\frac{1}{3}\right) + 3\left(\frac{1}{3}\right) = \boxed{2}.$$

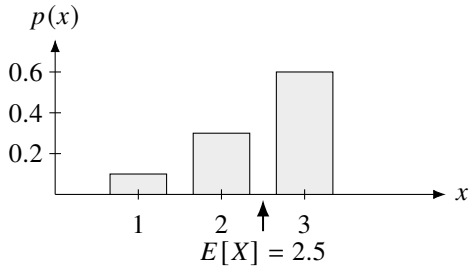


Example 2: Unequal Probabilities

Now let

$$\mathcal{X} = \{1, 2, 3\}, \qquad p(1) = 0.1, \quad p(2) = 0.3, \quad p(3) = 0.6.$$

The pmf places progressively more weight on the larger values:



Indeed,

$$E[X] = 1(0.1) + 2(0.3) + 3(0.6) = \boxed{2.5}.$$

Case 2: $h(x) = x^k$ **- Moments**

For $k = 1, 2, 3, \ldots,$

$$\boxed{E[X^k] = \sum_{x \in \mathcal{X}} x^k P(X = x).}$$

The quantity $E[X^k]$ is called the k**th moment** of X.

Case 3: $h(x) = ax + b$ **- Linearity**

Let a and b be constants. Then

$$\boxed{E[aX + b] = aE[X] + b.}$$

Therefore $E(\,\cdot\,)$ is a **linear operator**.

Proof. From $(*)$,

$$\begin{aligned} E[aX + b] &= \sum_{x \in \mathcal{X}} (ax + b) P(X = x) \\ &= a \sum_{x \in \mathcal{X}} x P(X = x) + b \sum_{x \in \mathcal{X}} P(X = x) \\ &= aE[X] + b(1) \\ &= \boxed{aE[X] + b}. \end{aligned}$$

Case 4: $h(x) = (x - \mu)^2$ **- Variance**

Let $\mu = E[X]$. The expected squared deviation from the mean,

$$E\left[(X - E[X])^2\right],$$

is called the **variance** of X. Thus

$$\boxed{V(X) = E[(X - \mu)^2] = \sum_{x \in \mathcal{X}} (x - \mu)^2 P(X = x).}$$

Variance is also denoted by

$$\boxed{V(X) = \sigma^2.}$$

The quantity σ is the **standard deviation** of X:

$$\boxed{\sigma = \text{sd}[X] = \sqrt{V(X)}.}$$

Standard deviation is used as a measure of the **variation** in the probability distribution of X.

Units and the Meaning of Standard Deviation

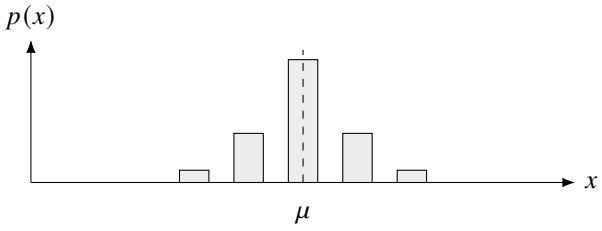
If X is measured in centimeters, then

$$X: \text{ cm}, \qquad E[X]: \text{ cm}, \qquad V(X): \text{ cm}^2, \qquad \text{sd}[X]: \text{ cm}.$$

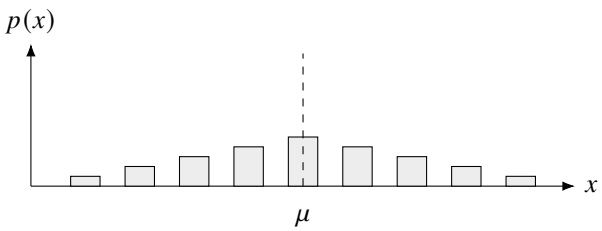
The standard deviation is therefore on the same scale as the original random variable.

Two probability histograms can have the same center μ but different spreads:

Narrower: X_1



Wider: X_2



$$\text{sd}[X_1] < \text{sd}[X_2].$$

Greater spread about a common mean corresponds to a larger standard deviation σ.

Example: Standard Deviation for an Unequal pmf

Return to the random variable with

$$\mathcal{X} = \{1, 2, 3\}, \qquad p(1) = 0.1, \quad p(2) = 0.3, \quad p(3) = 0.6.$$

Find $\text{sd}[X]$. Since $\mu = E[X] = 2.5$,

$$\begin{aligned}\sigma^2 = V(X) &= \sum_{x \in \mathcal{X}} (x - \mu)^2 P(X = x) \\ &= (1 - 2.5)^2(0.1) + (2 - 2.5)^2(0.3) + (3 - 2.5)^2(0.6).\end{aligned}$$

Finally, $\text{sd}[X] = \sqrt{V(X)}$.

Variance: An Alternative Formula

The variance can also be computed from the second moment and the mean:

$$\boxed{V(X) = E[X^2] - \big(E[X]\big)^2.}$$

That is, variance is the **second moment minus the square of the first moment**.

Proof. Since $\mu = E[X]$,

$$\begin{aligned}V(X) &= \sum_{x \in \mathcal{X}} (x - \mu)^2 P(X = x) \\ &= \sum_{x \in \mathcal{X}} (x^2 - 2\mu x + \mu^2) P(X = x) \\ &= \sum_{x \in \mathcal{X}} x^2 P(X = x) - 2\mu \sum_{x \in \mathcal{X}} x P(X = x) + \mu^2 \sum_{x \in \mathcal{X}} P(X = x) \\ &= E[X^2] - 2\mu \cdot \mu + \mu^2 \cdot 1 \\ &= E[X^2] - \mu^2 \\ &= \boxed{E[X^2] - \big(E[X]\big)^2}.\end{aligned}$$

Example: Using the Alternative Variance Formula

For $p(1) = 0.1$, $p(2) = 0.3$, and $p(3) = 0.6$, with $E[X] = 2.5$,

$$\begin{aligned}V(X) &= E[X^2] - \big(E[X]\big)^2 \\ &= \big[1^2(0.1) + 2^2(0.3) + 3^2(0.6)\big] - (2.5)^2 \\ &= [0.1 + 1.2 + 5.4] - 6.25 \\ &= 6.7 - 6.25 \\ &= \boxed{0.45} \quad \Longrightarrow \quad \text{sd}[X] = \boxed{\sqrt{0.45}}.\end{aligned}$$

Homework

Exercises: 2, 4, 6, 8, 10, 12, 14, 20, 22(a).

Named Discrete Distributions

Sections 3.1–3.3: Discrete random variables.

Sections 3.4–3.8: Named discrete probability distributions.

Section 3.4 - The Uniform and Geometric Distributions

I. Discrete Uniform Distribution

Let X take k possible values

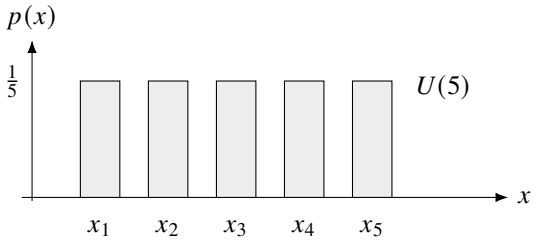
$$\mathcal{X} = \{x_1, x_2, \ldots, x_k\}.$$

If all k values are equally likely, then X has a **discrete uniform distribution** with parameter k, denoted

$$\boxed{X \sim U(k)}.$$

Thus

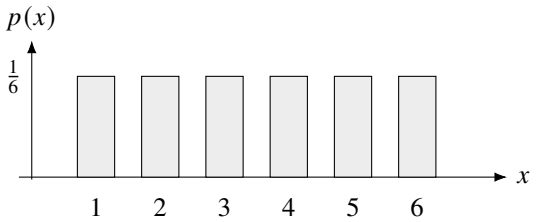
$$\boxed{P(X = x_i) = \frac{1}{k}, \qquad i = 1, 2, \ldots, k.}$$



Example: Rolling a Fair Six-Sided Die

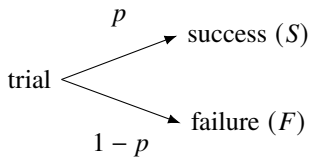
Roll a fair six-sided die and let X be the number of the side facing up. Then

$$\mathcal{X} = \{1, 2, 3, 4, 5, 6\}, \qquad \boxed{X \sim U(6)}, \qquad P(X = x) = \frac{1}{6} \ (x = 1, \ldots, 6).$$



II. The Geometric Distribution

Perform successive, identical, independent Bernoulli trials. Each trial has two outcomes:



Let

$$X = \text{the number of trials performed until the first success occurs.}$$

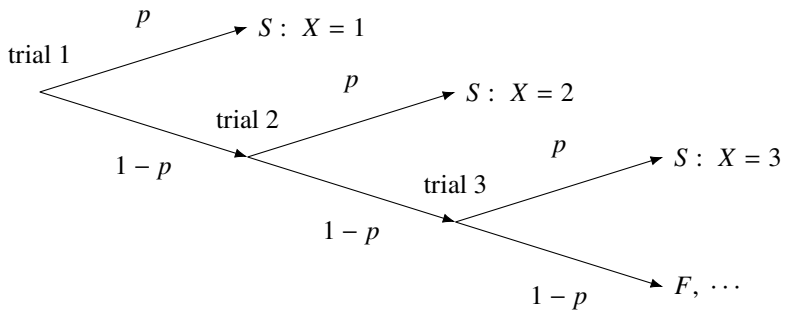
Then X has a **geometric distribution** with parameter p:

$$\boxed{X \sim G(p).}$$

The Geometric Stopping Tree and Support

The first success can occur on trial $1, 2, 3, \ldots$, so

$$\boxed{\mathcal{X} = \{1, 2, 3, 4, \ldots\}.}$$



Example 1: Cells Exposed Until Fusion

Let X be the number of cells exposed until fusion first occurs. A fusion is a success with probability $1/2$, while no fusion is a failure with probability $1/2$. Thus

$$\boxed{X \sim G\left(\frac{1}{2}\right).}$$

$X = 1$ when the first cell fuses, $X = 2$ when the second is the first to fuse, and so on.

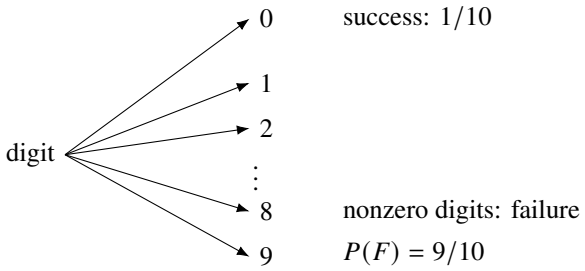
Example 2: Generating Random Digits Until the First Zero

Generate digits independently from $0, 1, 2, \ldots, 9$, each with probability $1/10$. Let

$$X = \text{the number of generated digits until the first zero appears.}$$

A zero counts as a success, so $p = 1/10$ and

$$\boxed{X \sim G\left(\frac{1}{10}\right)}.$$



Two possible digit sequences are

$$\begin{aligned} 371178880 &\quad\Longrightarrow\quad X = 9, \\ 51552270 &\quad\Longrightarrow\quad X = 8. \end{aligned}$$

Derivation of the Geometric pmf

Write T_j for the result of trial j. For $x = 1, 2, 3, \ldots$, the event $X = x$ means that the first $x - 1$ trials are failures and trial x is a success:

$$\begin{aligned} p(x) = P(X = x) &= P(T_1 = F,\ T_2 = F,\ \ldots,\ T_{x-1} = F,\ T_x = S) \\ &= P(T_1 = F) \cdots P(T_{x-1} = F) P(T_x = S) \qquad \text{(independence)} \\ &= \underbrace{(1-p) \cdots (1-p)}_{x-1 \text{ factors}}\ p \\ &= \boxed{(1-p)^{x-1} p}. \end{aligned}$$

Cumulative Distribution Function of a Geometric Variable

For $X \sim G(p)$, the cdf is zero for $x < 1$. For $x \geq 1$,

$$P(X \leq x) = 1 - P(X > x) = 1 - (1-p)^{\lfloor x \rfloor},$$

where $\lfloor x \rfloor$ denotes the greatest integer not exceeding x. Hence

$$\boxed{F(x) = \begin{cases} 0, & x < 1, \\ 1 - (1-p)^{\lfloor x \rfloor}, & x \geq 1. \end{cases}}$$

For the cell-fusion example, $X \sim G(\frac{1}{2})$:

$$\boxed{P(X \leq 17) = 1 - \left(\frac{1}{2}\right)^{17}.}$$

Mean and Interpretation of a Geometric Variable

Property. If $X \sim G(p)$, then

$$\boxed{E[X] = \frac{1}{p}.}$$

For the random-digit experiment, $p = 1/10$, so

$$\boxed{E[X] = \frac{1}{1/10} = 10.}$$

The mean number of digits until the first zero is 10. Possible observed values include

$$X_1 = 8, \quad X_2 = 9, \quad X_3 = 16, \quad X_4 = 8, \quad X_5 = 10, \quad X_6 = 7, \quad \ldots.$$

For instance, 47411150 gives $X_1 = 8$, and 616582230 gives $X_2 = 9$.

Standard Deviation of a Geometric Variable

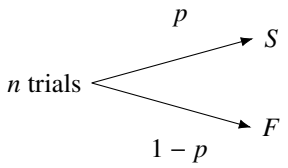
Property. If $X \sim G(p)$, then

$$\boxed{\mathrm{sd}[X] = \frac{\sqrt{1-p}}{p}.}$$

Section 3.5 - The Binomial Distribution

I. Definition

Perform n independent Bernoulli trials:



Let

$$X = \text{the number of successes in the } n \text{ trials.}$$

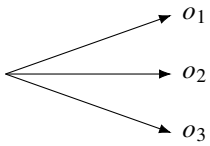
Then X has the **binomial distribution** with parameters n and p, denoted

$$\boxed{X \sim B(n, p).}$$

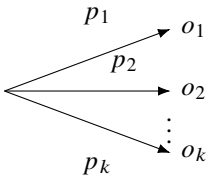
Two Outcomes versus Several Outcomes

A trial with three possible outcomes o_1, o_2, o_3 gives a **trinomial distribution**. With k outcomes $o_1, \ldots, o_k$ and probabilities $p_1, \ldots, p_k$, the distribution is **multinomial**.

Trinomial



Multinomial



Geometric versus Binomial Experiments

Distribution	Number of trials	Number of successes
$G(p)$	X (random)	1
$B(n, p)$	n (fixed)	X (random)

Example. Generate 25 independent random digits from 0 through 9. Let

$$X = \text{the number of zeros among the 25 digits.}$$

Since zero has probability $1/10$ on every trial,

$$\boxed{X \sim B\left(25, \frac{1}{10}\right).}$$

Probability Mass Function of the Binomial Distribution

For $X \sim B(n, p)$, one sequence with x successes and $n - x$ failures has probability

$$p^x(1-p)^{n-x}.$$

There are $\binom{n}{x}$ possible orders. Thus

$$\boxed{P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \ldots, n.}$$

The **binomial coefficient** (Section 1.3) is $\binom{n}{x} = \dfrac{n!}{x!(n-x)!}$.

Example: Exactly Three Zeros Among 25 Random Digits

For $X \sim B(25, \frac{1}{10})$, the probability of exactly three zeros is

$$\boxed{P(X = 3) = \binom{25}{3} (0.1)^3 (0.9)^{22}.}$$

The binomial coefficient is

$$\binom{25}{3} = \frac{25!}{3!\,22!} = \frac{25 \cdot 24 \cdot 23}{3 \cdot 2 \cdot 1}.$$